

1 Cell is called the _____ of life.

(a) Tissue

(a) Organ

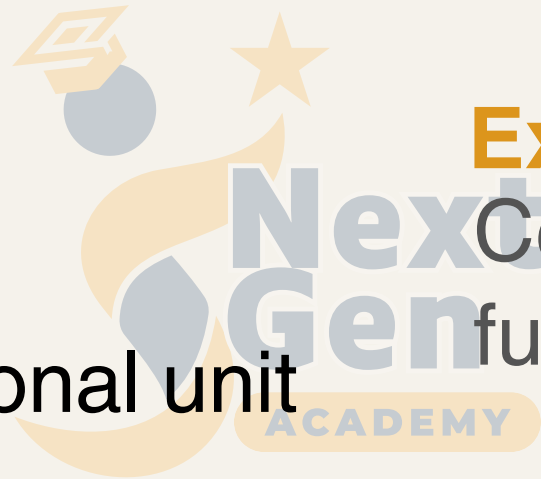
(a) Structural and functional unit

(a) Nucleus

Explanation:

Cell is the structural and functional unit of life.

Ans: C



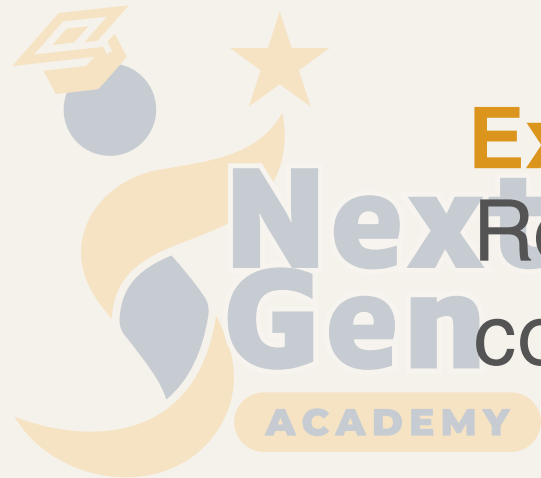
2 Who discovered the cell in 1665?

- (a) Schleiden
- (a) Schwann
- (a) Robert Hooke
- (a) Virchow

Explanation:

Robert Hooke observed cork cells.

Ans: C



3 The smallest cell is found in:

- (a) Bacteria
- (a) Mycoplasma
- (a) RBC
- (a) Virus

Explanation:

Mycoplasma is the smallest living cell.

Ans: B



4 Which organelle is called the “Powerhouse of the cell”?

- (a) Nucleus
- (a) Chloroplast
- (a) Mitochondria
- (a) Golgi body

Explanation:

Mitochondria produce ATP.



Ans: C

5 Prokaryotic cells lack:

- (a) DNA
- (a) Ribosomes
- (a) Nucleus
- (a) Cytoplasm

Explanation:

Prokaryotes have no true nucleus.

Ans: C



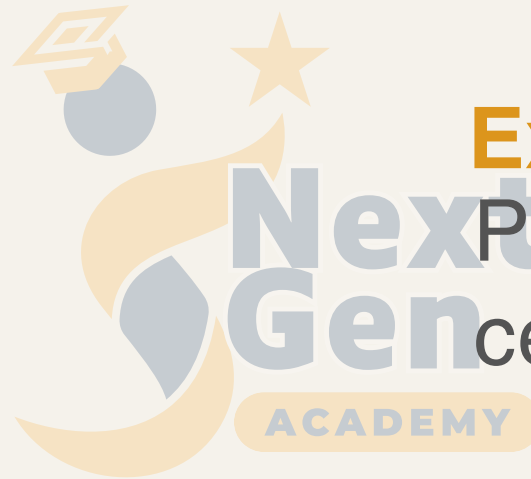
6 Cell wall in plants is made of:

- (a) Chitin
- (a) Cellulose
- (a) Peptidoglycan
- (a) Protein

Explanation:

Plant cell wall is made of cellulose.

Ans: B



7 Which cell organelle is double-membrane bound?

- (a) Ribosome
- (a) Mitochondria
- (a) Lysosome
- (a) Golgi body

Explanation:

Mitochondria have double membranes.

Ans: B



8 Fluid Mosaic Model was proposed by:

- (a) Singer and Nicolson
- (a) Watson and Crick
- (a) Schleiden and Schwann
- (a) Robert Brown

Explanation:

Singer and Nicolson proposed the Fluid Mosaic Model.

Ans: A

9 Who discovered nucleus?

- (a) Robert Hooke
- (a) Robert Brown
- (a) Virchow
- (a) Schwann

Explanation:

Robert Brown discovered nucleus in orchid cells.

Ans: B



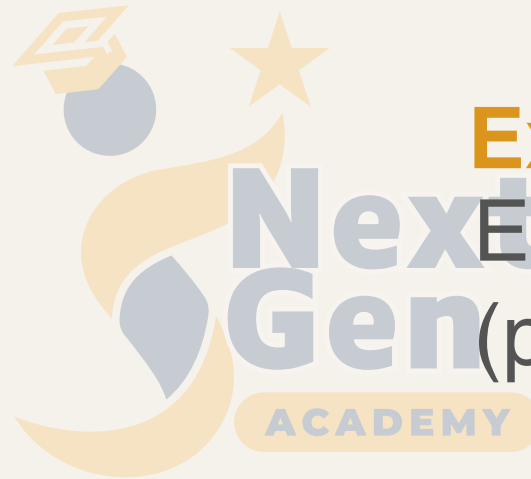
10 Which is a prokaryote?

- (a) Amoeba
- (a) Paramecium
- (a) E. coli
- (a) Hydra

Explanation:

E. coli is a bacterium (prokaryote).

Ans: C



11 Ribosomes are made up of:

- (a) Protein + DNA
- (a) RNA + Protein
- (a) DNA + RNA
- (a) Lipids

Explanation:

Ribosomes are made of rRNA + proteins.

Ans: B



12 Largest cell in human body:

- (a) Nerve cell
- (a) RBC
- (a) Liver cell
- (a) Ovum

Explanation:

Human ovum is the largest cell.

Ans: D



13 Cell theory was proposed by:

- (a) Hooke & Brown
- (a) Schleiden & Schwann
- (a) Darwin & Wallace
- (a) Watson & Crick

Explanation:

Schleiden and Schwann proposed cell theory.

Ans: B



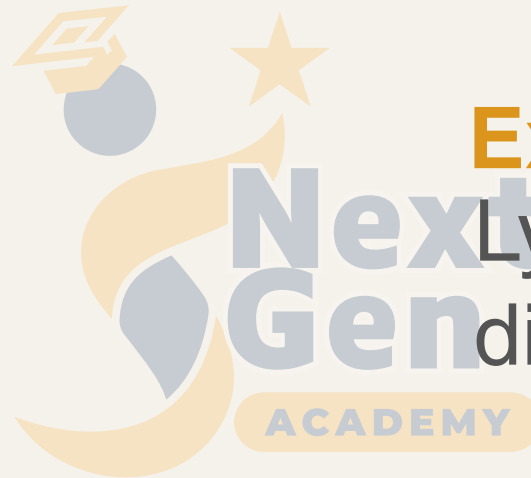
14 Which organelle is called “Suicidal bag”?

- (a) Ribosome
- (a) Lysosome
- (a) Golgi body
- (a) Vacuole

Explanation:

Lysosomes contain digestive enzymes.

Ans: B



15 In plants, energy is stored in:

- (a) Starch
- (a) Glycogen
- (a) Cellulose
- (a) Protein

Explanation:

Plants store energy as starch.

Ans: A



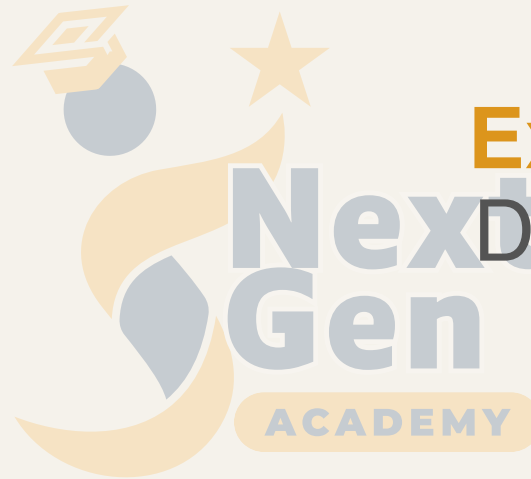
16 DNA is located in:

- (a) Mitochondria only
- (a) Chloroplast only
- (a) Nucleus (mainly)
- (a) Lysosome

Explanation:

DNA is mainly in nucleus.

Ans: C



17 Which of the following is absent in animal cells?

- (a) Ribosome
- (a) Cell wall
- (a) Nucleus
- (a) Mitochondria

Explanation:

Animal cells lack cell walls.

Ans: B



18 What is the size of a typical prokaryotic cell?

- (a) 1–10 μm
- (a) 10–100 μm
- (a) 100–1000 μm
- (a) 0.01 μm

Explanation:

Prokaryotic cells are small (1–10 μm).

Ans: A



19 Which structure controls cell activities?

- (a) Ribosome
- (a) Nucleus
- (a) Lysosome
- (a) ER

Explanation:

Nucleus controls cell activities.

Ans: B



20 Chloroplasts contain:

- (a) DNA & Ribosomes
- (a) Only DNA
- (a) Only proteins
- (a) Only RNA

Explanation:

Chloroplasts have DNA & ribosomes (semi-autonomous).

Ans: A



21 Which organelle packages proteins?

- (a) ER
- (a) Golgi body
- (a) Lysosome
- (a) Mitochondria

Explanation:

Golgi apparatus modifies and packages proteins.

Ans: B



22 Who gave the concept “Omnis cellula e cellula”?

- (a) Virchow
- (a) Hooke
- (a) Schwann
- (a) Brown

Explanation:

Virchow stated “Cells arise from pre-existing cells.”

Ans: A



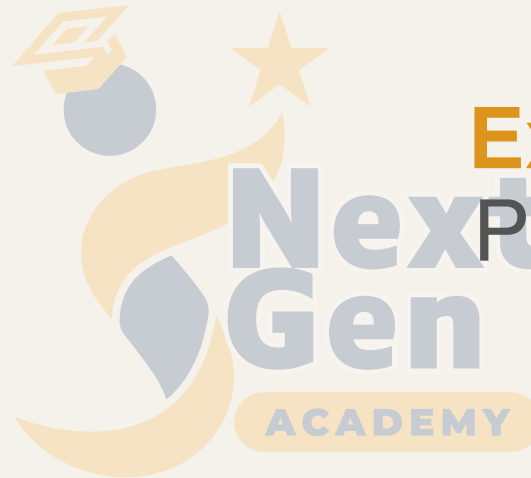
23 Centrioles are absent in:

- (a) Plant cells
- (a) Animal cells
- (a) Both
- (a) None

Explanation:

Plant cells lack centrioles.

Ans: A



24 Which organelle is called “Protein factory”?

- (a) Lysosome
- (a) Ribosome
- (a) Mitochondria
- (a) Golgi body



Explanation:

Ribosomes synthesize proteins.

Ans: B

25 The main function of smooth ER:

- (a) Protein synthesis
- (a) Lipid synthesis
- (a) ATP synthesis
- (a) Digestion

Explanation:

Smooth ER synthesizes lipids.

Ans: B



26 Which type of ribosome is found in prokaryotes?

(a) 80S

(a) 70S

(a) 60S

(a) 40S

Explanation:

Prokaryotes have 70S ribosomes.

Ans: B



27 Cristae are present in:

- (a) Lysosome
- (a) Golgi body
- (a) Mitochondria
- (a) ER

Ans: C



Explanation:

Inner folds of mitochondria are cristae.

28 Chromatin condenses to form:

- (a) Cytoplasm
- (a) Nucleolus
- (a) Chromosomes
- (a) Ribosomes

Explanation:

Chromatin →
Chromosomes during
division.

Ans: C



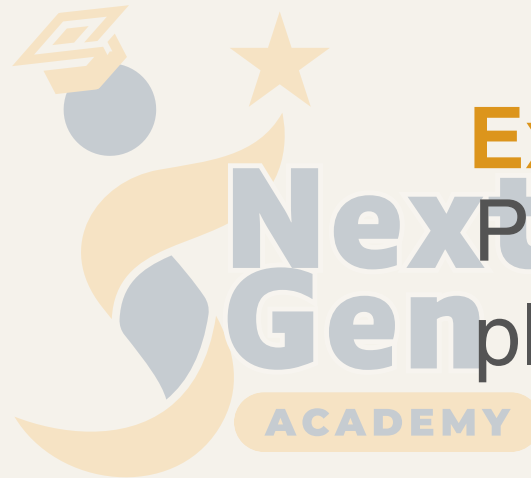
29 Plasmodesmata are connections between:

- (a) Animal cells
- (a) Plant cells
- (a) Bacteria
- (a) Viruses

Explanation:

Plasmodesmata connect plant cells.

Ans: B



30 Tonoplast is:

- (a) Mitochondrial membrane
- (a) Nuclear membrane
- (a) Vacuolar membrane
- (a) Ribosomal cover

Explanation:

Tonoplast is the membrane of vacuole.

Ans: C



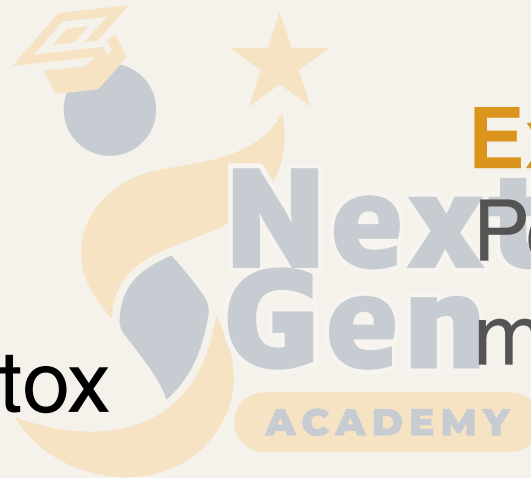
31 Peroxisomes are involved in:

- (a) Photosynthesis
- (a) Respiration
- (a) Fat metabolism & detox
- (a) Protein synthesis

Explanation:

Peroxisomes detoxify and metabolize fats.

Ans: C



32 The site of ribosome assembly:

- (a) ER
- (a) Nucleolus
- (a) Golgi
- (a) Cytoplasm

Explanation:

Ribosomes are assembled in nucleolus.

Ans: B



33 Bacterial cell wall is made of:

- (a) Cellulose
- (a) Peptidoglycan
- (a) Protein
- (a) Pectin

Explanation:

Bacterial wall =
peptidoglycan.

Ans: B



34 The universal energy currency of cell:

(a) ADP

(a) ATP

(a) AMP

(a) GTP

Explanation:

ATP is universal energy currency.

Ans: B



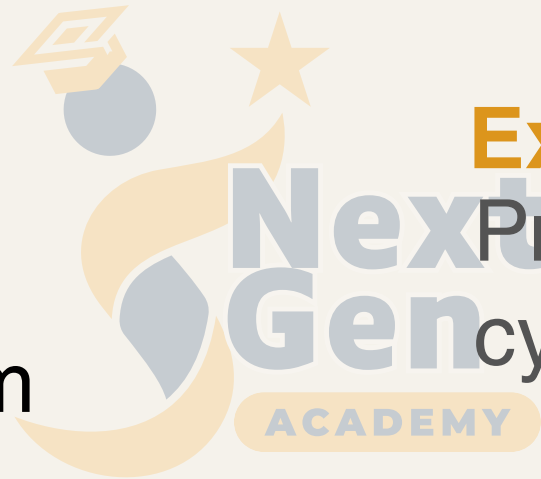
35 Protoplasm is:

- (a) Nucleus
- (a) Cytoplasm
- (a) Nucleus + Cytoplasm
- (a) Ribosome

Explanation:

Protoplasm = nucleus + cytoplasm.

Ans: C



36 Middle lamella in plant cells is rich in:

- (a) Cellulose
- (a) Lignin
- (a) Pectin
- (a) Chitin

Explanation:

Middle lamella is rich in calcium pectate.

Ans: C



37 Mitochondria and chloroplasts are called:

- (a) Endomembrane system
- (a) Semi-autonomous organelles
- (a) Lysosomes
- (a) Ribosomes

Explanation:

They contain DNA, RNA, ribosomes, and divide.

Ans: B

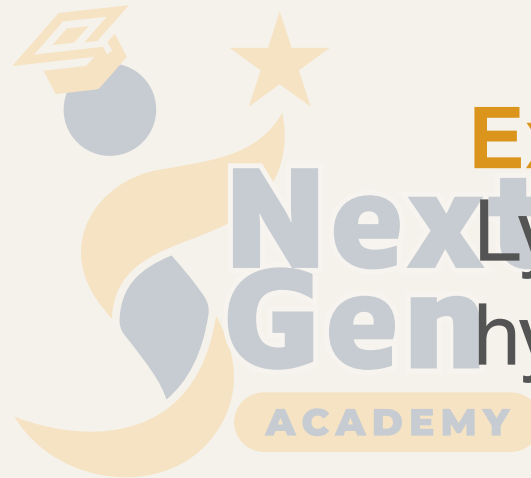
38 Lysosomes are rich in:

- (a) Hydrolase enzymes
- (a) Oxidase enzymes
- (a) Transferases
- (a) Ligases

Explanation:

Lysosomes have hydrolytic enzymes.

Ans: A



39 9+2 arrangement is found in:

- (a) Ribosome
- (a) Cilia & Flagella
- (a) Centriole
- (a) Spindle fibres

Explanation:

Cilia/flagella have 9+2 microtubule arrangement.

Ans: B



40 Eukaryotic cells evolved through:

- (a) Endosymbiotic theory
- (a) Abiogenesis
- (a) Biogenesis
- (a) Spontaneous generation

Explanation:

Endosymbiotic theory explains organelle origin.

Ans: A



41 Ribophorins are found in:

- (a) SER
- (a) RER
- (a) Mitochondria
- (a) Golgi

Explanation:

Ribophorins are proteins in RER.

Ans: B



42 Mesosome is present in:

- (a) Fungi
- (a) Protozoa
- (a) Bacteria
- (a) Virus

Explanation:

Mesosome = infoldings in bacterial membrane.

Ans: C



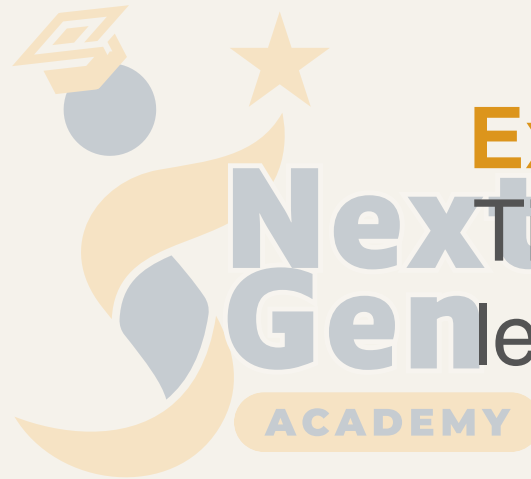
43 Which cell junction prevents leakage?

- (a) Desmosome
- (a) Tight junction
- (a) Gap junction
- (a) Plasmodesmata

Explanation:

Tight junctions prevent leakage.

Ans: B



44 Glycocalyx is:

- (a) Protein coat
- (a) Lipid coat
- (a) Carbohydrate-rich covering
- (a) DNA coat

Explanation:

Glycocalyx is carbohydrate-rich layer.

Ans: C



45 Microtubules are made of:

- (a) Tubulin
- (a) Actin
- (a) Myosin
- (a) Collagen

Explanation:

Microtubules are composed of tubulin.

Ans: A



46 Which organelle shows polymorphism?

- (a) ER
- (a) Ribosome
- (a) Golgi body
- (a) Mitochondria

Explanation:

Golgi shows structural polymorphism.

Ans: C



47 Which organelle performs photorespiration?

- (a) Peroxisome
- (a) Lysosome
- (a) Ribosome
- (a) ER

Explanation:

Peroxisomes participate in photorespiration.

Ans: A



48 WBCs engulf bacteria by:

- (a) Pinocytosis
- (a) Exocytosis
- (a) Phagocytosis
- (a) Endosmosis

Explanation:

Phagocytosis = cell eating.

Ans: C



49 The spindle apparatus is made of:

- (a) Microfilaments
- (a) Microtubules
- (a) Intermediate filaments
- (a) Actin

Explanation:

Spindle fibers are microtubules.

Ans: B



50

Which cell organelle is rich in hydrolytic enzymes and important in apoptosis?

- (a) Ribosome
- (a) Golgi
- (a) Lysosome
- (a) Nucleolus

Explanation:

Lysosomes mediate autolysis (apoptosis).

Ans: C

